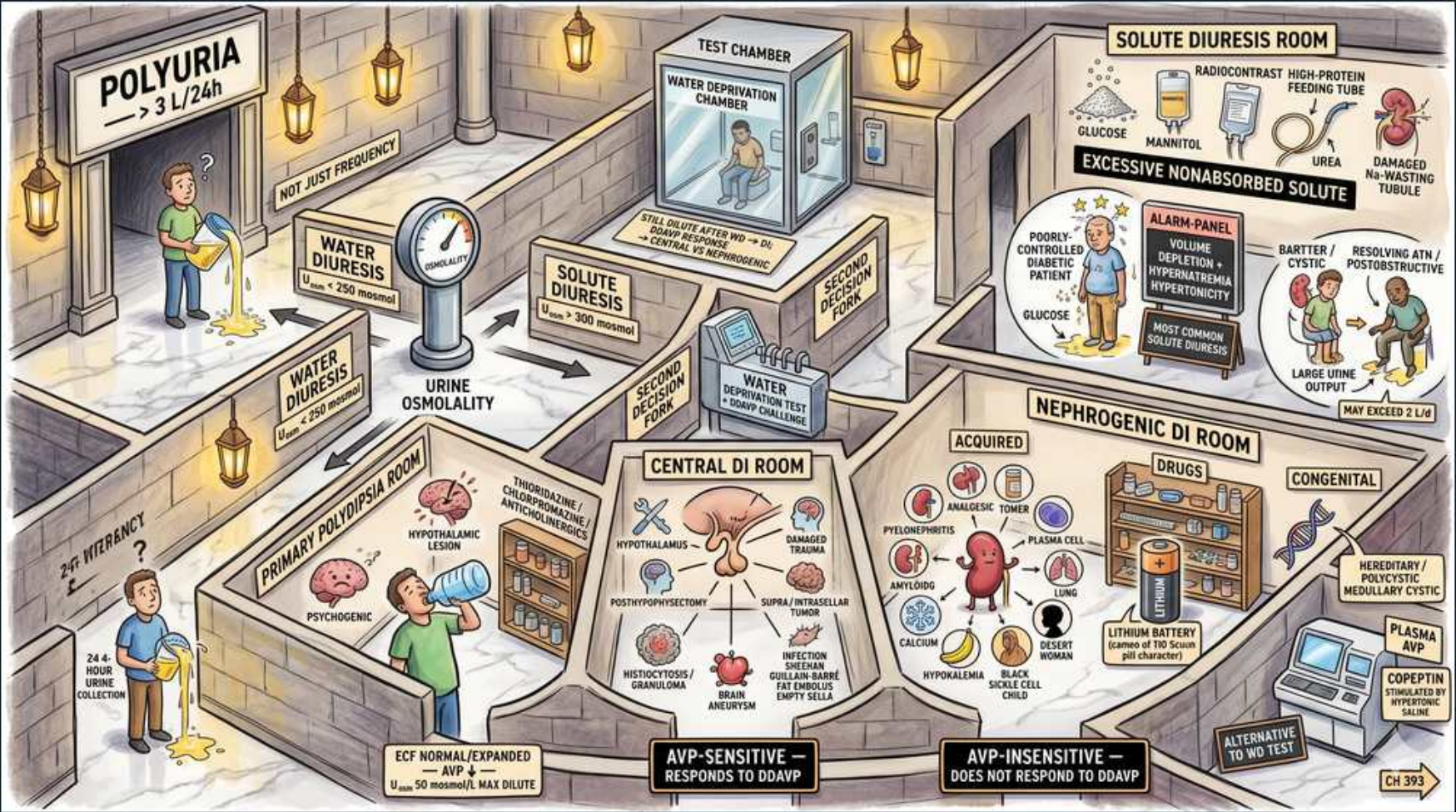


Renal Approach — Polyuria (osmotic vs water diuresis, DI types, water deprivation test)



1. Polyuria definition >3 L/24h

WHAT YOU SEE

Top-left doorway sign POLYURIA >3 L/24h

WHAT IT ENCODES

Polyuria is urine output >3 L/day (or ~>40-50 mL/kg/day) confirmed by a 24-hour collection. The first step is to confirm true polyuria and separate it from urinary FREQUENCY (frequent small voids, e.g., cystitis, BPH, overactive bladder) and from nocturia. Once confirmed, the master branch point is urine osmolality, which splits polyuria into water diuresis (dilute urine) vs solute/osmotic diuresis (concentrated urine). Mnemonic — 'confirm >3 L, then read the urine osmolality.'

BOARD TRAP

Tempting wrong answer: 'patient urinates many times a day, so this is polyuria — work up diabetes insipidus.' Discriminator: urinary FREQUENCY is many small-volume voids (irritative/obstructive bladder pathology) with a normal or low 24h total, NOT polyuria. Quantify the 24h volume first; only >3 L/day enters the polyuria algorithm.

2. Urine osmolality fork

WHAT YOU SEE

Central gauge URINE OSMOLALITY directing water-diuresis vs solute-diuresis paths

WHAT IT ENCODES

Urine osmolality is the key fork. A LOW urine osm (<300, roughly <250-300 mOsm/kg) means free water is being excreted = WATER diuresis (central DI, nephrogenic DI, primary polydipsia). A HIGH urine osm (>300, classically with daily solute excretion >1000-1200 mOsm) means an osmotic solute is dragging water out = SOLUTE/OSMOTIC diuresis. Total daily solute output (Uosm × volume) confirms which side you're on. Mnemonic — 'dilute = water problem; concentrated = solute problem.'

BOARD TRAP

Tempting wrong answer: 'polyuria with a high urine osmolality is diabetes insipidus.' Discriminator: DI produces DILUTE urine (low Uosm) because ADH action is absent; a HIGH urine osm with polyuria indicates OSMOTIC diuresis (glucose, urea, mannitol), so chase hyperglycemia/feeds, not DI.

3. Water diuresis branch

WHAT YOU SEE

Left arrow WATER DIURESIS — dilute urine

WHAT IT ENCODES

Water diuresis = excretion of large volumes of DILUTE urine (low Uosm) and has three causes: central diabetes insipidus (deficient ADH), nephrogenic DI (ADH-resistant kidney), and primary polydipsia (excess water intake suppresses ADH appropriately). Serum sodium helps triage them: the DIs tend toward HIGH-normal/elevated Na (water loss), whereas primary polydipsia trends LOW-normal/low Na (water excess). Mnemonic — 'dilute urine = the 3 P/D players: Pituitary (central), Pipe (nephrogenic), Polydipsia.'

BOARD TRAP

Tempting wrong answer: 'all dilute-urine polyuria is diabetes insipidus.' Discriminator: primary polydipsia ALSO gives dilute polyuria but the ADH suppression is APPROPRIATE (response to water loading) and serum Na is low/low-normal, opposite the high Na of DI. The water deprivation test separates them.

4. Solute / osmotic diuresis branch

WHAT YOU SEE

Right arrow SOLUTE DIURESIS toward solute rooms

WHAT IT ENCODES

Solute (osmotic) diuresis occurs when a non-reabsorbed osmotically active substance in the tubular lumen holds water and drags it into the urine, producing high-osmolality polyuria. Classic drivers: glucose (uncontrolled diabetes/hyperglycemic crises — the most common cause; SGLT2 inhibitors), urea (high-protein tube feeds, recovery from AKI/post-obstruction), and mannitol/contrast/IV saline loads. Mnemonic — 'glucose, urea, mannitol — the osmotic trio.'

BOARD TRAP

Tempting wrong answer: 'polyuria in a diabetic with glucose 450 means they developed diabetes insipidus.' Discriminator: hyperglycemia causes OSMOTIC (glucose) diuresis with a HIGH urine osm — treat the hyperglycemia. DI is a WATER diuresis with dilute urine. Don't conflate diabetes MELLITUS (glucose osmotic) with diabetes INSIPIDUS (ADH problem).

5. Solute diuresis room — glucose/urea

WHAT YOU SEE

SOLUTE DIURESIS ROOM: glucose IV bag, urea, high-protein feeding tube, damaged Na-wasting tubule

WHAT IT ENCODES

Inside the osmotic-diuresis room the drivers are: hyperglycemia (glucosuria once plasma glucose exceeds the ~180 mg/dL renal threshold), urea load (high-protein enteral feeds, hypercatabolism), exogenous mannitol/contrast, and salt-wasting from a damaged tubule (recovering ATN, post-obstructive diuresis). Each leaves osmoles in the lumen that water must follow. Mnemonic — 'sugar, protein-feeds, mannitol, and a leaky salt tubule.'

BOARD TRAP

Tempting wrong answer: 'polyuria after relief of urinary obstruction or during ATN recovery is the patient drinking too much.' Discriminator: post-obstructive and ATN-recovery diuresis is an intrinsic SOLUTE/salt-and-urea diuresis from tubular dysfunction, not primary polydipsia — it is largely self-limited but requires monitoring for volume depletion and electrolyte losses, not fluid restriction.

6. Solute alarm panel

WHAT YOU SEE

ALARM PANEL: volume depletion, hypernatremia, hyperosmolality; resolving post-obstructive

WHAT IT ENCODES

Sustained osmotic diuresis drives loss of water in excess of solute, producing volume depletion, hypernatremia, and a high plasma osmolality (the hyperglycemic hyperosmolar state is the extreme example). Post-obstructive diuresis is usually appropriate/self-limited (excreting retained solute and water) but can occasionally be pathologic and cause dangerous depletion, so monitor volume and electrolytes and replace judiciously. Mnemonic — 'osmotic diuresis dries you out and salts you up.'

BOARD TRAP

Tempting wrong answer: 'salt-wasting polyuria with metabolic ALKALOSIS and hypokalemia is just osmotic diuresis.' Discriminator: hypokalemic metabolic ALKALOSIS with renal salt wasting points to Bartter (loop-like) or Gitelman (thiazide-like) syndromes — distinct tubulopathies — rather than a simple osmotic/post-obstructive diuresis. Check potassium, acid-base, and urine electrolytes.

7. Primary polydipsia room

WHAT YOU SEE

PRIMARY POLYDIPSIA ROOM: person drinking, AVP↓, ECF normal/expanded, hyponatremia/low Na

WHAT IT ENCODES

Primary (psychogenic/dipsogenic) polydipsia is compulsive or habitual excessive water intake that appropriately SUPPRESSES AVP/ADH; the kidney correctly excretes the water load, so ECF is normal/expanded and serum sodium runs LOW or low-normal. Seen in psychiatric illness (schizophrenia), and some medications/dry-mouth states. Chronic high flow can wash out the medullary gradient, blunting maximal concentrating ability even after deprivation. Mnemonic — 'drinks too much → low ADH, low Na, dilute urine.'

BOARD TRAP

Tempting wrong answer: 'dilute polyuria — diagnose central DI and start desmopressin.' Discriminator: giving DDAVP to a primary polydipsic who keeps drinking causes dangerous water retention and HYPONATREMIA. The low/low-normal serum Na (vs the high Na of DI), appropriate ADH suppression, and partial concentration with water deprivation alone distinguish polydipsia from DI.

8. Central DI room

WHAT YOU SEE

CENTRAL DI ROOM: brain, AVP-deficient causes (trauma, post-surgery, tumor, infiltration, ischemia)

WHAT IT ENCODES

Central DI is DEFICIENT production/release of ADH (AVP) from the hypothalamus/posterior pituitary, giving dilute polyuria with a tendency to hypernatremia. Causes: idiopathic/autoimmune, pituitary surgery or head trauma, tumors (craniopharyngioma, metastasis), infiltration (sarcoidosis, Langerhans cell histiocytosis), Sheehan syndrome, and hypoxic/ischemic injury. Treatment is desmopressin (DDAVP). Mnemonic — central = 'brain doesn't make ADH.'

BOARD TRAP

Tempting wrong answer: 'central and nephrogenic DI are treated the same way with DDAVP.' Discriminator: central DI RESPONDS to desmopressin (urine concentrates), nephrogenic does NOT. The DDAVP response after water deprivation is the defining test; treating nephrogenic DI with DDAVP fails.

9. AVP-sensitive responds to DDAVP

WHAT YOU SEE

Banner AVP-SENSITIVE — RESPONDS TO DDAVP under central DI

WHAT IT ENCODES

Central DI is AVP-SENSITIVE: after the water deprivation test, giving desmopressin raises urine osmolality SUBSTANTIALY (classically by >50%, or >9-15%), confirming the kidney can respond and the defect is central ADH deficiency. Long-term therapy is intranasal/oral DDAVP titrated to avoid hyponatremia. Mnemonic — 'central = sensitive to the synthetic ADH.'

BOARD TRAP

Tempting wrong answer: 'urine concentrated after DDAVP, so this must be nephrogenic DI.' Discriminator: a robust rise in urine osmolality after DDAVP = CENTRAL DI (kidney responds to replaced hormone). Nephrogenic DI shows little/no rise (<~9-15%) because the receptor/aquaporin defect prevents the kidney from responding.

10. Nephrogenic DI — acquired

WHAT YOU SEE

NEPHROGENIC DI ROOM acquired: drugs, plasma cell, hypercalcemia, hypokalemia

WHAT IT ENCODES

Acquired nephrogenic DI = the kidney cannot respond to ADH. Leading causes: lithium (most common drug cause — impairs collecting-duct V2 signaling/aquaporin-2), chronic HYPERCALCEMIA and chronic HYPOKALEMIA (both impair concentrating ability), and chronic kidney/tubulointerstitial disease; also demeclocycline, foscarnet, and post-obstructive states. Treat the cause and use thiazides + low-salt/low-protein diet ± amiloride (especially for lithium) to reduce urine volume. Mnemonic — 'Lithium, hyperCalcemia, hypoKalemia poison the pipe.'

BOARD TRAP

Tempting wrong answer: 'lithium-treated bipolar patient with dilute polyuria has central DI — give DDAVP.' Discriminator: lithium causes NEPHROGENIC DI (ADH-resistant) which does NOT respond to DDAVP; manage by minimizing lithium, correcting Ca/K, and adding a thiazide ± amiloride, not by giving desmopressin.

11. Nephrogenic DI — congenital

WHAT YOU SEE

CONGENITAL DNA/helix: hereditary, polycystic, medullary cystic

WHAT IT ENCODES

Congenital nephrogenic DI presents in infancy/childhood with polyuria, failure to thrive, and hypernatremic dehydration. ~90% is X-linked from AVPR2 (V2 receptor) loss-of-function mutations (affects males); the remainder is autosomal from AQP2 (aquaporin-2 water channel) mutations. Hereditary cystic diseases (medullary cystic/nephronophthisis) can also impair concentrating ability. Mnemonic — 'born with a broken V2 receptor or aquaporin-2.'

BOARD TRAP

Tempting wrong answer: 'an infant boy with hypernatremic dehydration and dilute urine — central DI from a brain lesion.' Discriminator: X-linked congenital nephrogenic DI (AVPR2 mutation) is the classic cause in a male infant and does NOT respond to DDAVP; central DI would respond. Genetics, family history, and the DDAVP non-response point to the receptor defect.

12. Lithium battery cue

WHAT YOU SEE

Lithium battery icon in nephrogenic DI room

WHAT IT ENCODES

Lithium enters collecting-duct principal cells via ENaC and disrupts V2-receptor signaling and aquaporin-2 expression, producing acquired nephrogenic DI in up to ~20-40% of chronic users. It can persist even after stopping the drug. Amiloride (blocks ENaC, limiting lithium entry) plus thiazides reduces polyuria; monitor lithium levels and renal function. Mnemonic — 'lithium charges up the urine output.'

BOARD TRAP

Tempting wrong answer: 'add a loop diuretic to control lithium-induced polyuria.' Discriminator: the evidence-based agents are THIAZIDES (paradoxically reduce urine volume via mild volume contraction and enhanced proximal reabsorption) plus AMILORIDE (blocks lithium entry through ENaC) — loop diuretics worsen free-water loss and are not the answer here.

13. AVP-insensitive — no DDAVP response

WHAT YOU SEE

Banner AVP-INSENSITIVE — DOES NOT RESPOND TO DDAVP

WHAT IT ENCODES

Nephrogenic DI is AVP-INSENSITIVE: after water deprivation, urine osmolality stays inappropriately LOW and rises minimally (<~9-15%) even AFTER desmopressin, because the collecting duct cannot respond despite adequate (often elevated) endogenous ADH/copeptin. This non-response is the defining feature separating it from central DI.

Mnemonic — 'kidney is deaf to ADH, so DDAVP changes nothing.'

BOARD TRAP

Tempting wrong answer: 'no rise in urine osm after DDAVP means the test failed — repeat it.' Discriminator: a flat urine osmolality after DDAVP is the diagnostic finding of NEPHROGENIC DI (kidney can't respond), not a technical failure. Pair it with elevated endogenous ADH/copeptin to confirm.

14. Water deprivation test / copeptin

WHAT YOU SEE

Top WATER DEPRIVATION CHAMBER with seated patient; COPEPTIN alternative noted at right

WHAT IT ENCODES

The water deprivation (Miller) test: withhold fluids under supervision, measure serial urine and plasma osmolality until plasma osm/Na rises or urine osm plateaus, then give desmopressin. Interpretation — normal/primary polydipsia concentrate WITH deprivation alone (Uosm rises adequately); central DI fails to concentrate with deprivation but RESPONDS to DDAVP (Uosm rises >50%); nephrogenic DI fails BOTH. Hypertonic-saline–stimulated COPEPTIN (a stable AVP surrogate) is a faster, more accurate modern alternative. Mnemonic — 'deprive, then DDAVP: polydipsia passes, central answers to DDAVP, nephrogenic fails both.'

BOARD TRAP

Tempting wrong answer: 'patient concentrated urine after fluid restriction, so they don't have DI — it's all behavioral.' Discriminator: PRIMARY POLYDIPSIA concentrates with deprivation alone (no DDAVP needed) and is the benign mimic, but PARTIAL central or partial nephrogenic DI can also partially concentrate — interpret the magnitude of rise and the post-DDAVP response (or copeptin) rather than assuming any concentration excludes DI.